

PyGameMaker — Tutorial 10: Turn-Based Multiplayer by File Exchange

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Turn-Based Multiplayer: Tic-Tac-Toe by File Exchange** (5 pages, about 25-30 minutes). Keep the tutorial open on one half of your screen; it has the full details of every step. Tick each box when the step is done, and check the **You should see** line at the end of every phase. From Phase 2 you need a partner and a **shared folder** that both computers can use. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

Tic-Tac-Toe for two players on two computers, with no live connection: each move is written as a small file in a shared folder, and the other computer reads it. One computer **hosts** (press H) and plays X; the other **joins** (press J) and plays O.

Phase 1: The Board

- ☐ **1.** Create a project TicTacToeFiles with the **Empty Project** template. Create obj_game (no sprite) and a room room_board, and place one obj_game in it.
- ☐ **2.** In obj_game, **Create** event: set nine variables to an empty text: cell_0_0, cell_1_0, cell_2_0, cell_0_1, cell_1_1, cell_2_1, cell_0_2, cell_1_2, cell_2_2 (column first, then row).
- ☐ **3. Draw** event: first **Set draw color** white. Then four **Draw line** actions for the grid: x 270 and 370 from y 90 to 390, and y 190 and 290 from x 170 to 470.
- ☐ **4. Mouse: Left Button Pressed:** nine **If** checks, one per cell, like cell_0_0 == "" and mouse_x >= 170 and mouse_x < 270 and mouse_y >= 90 and mouse_y < 190 then **Set variable** cell_0_0 to X.
- ☐ **5.** In the **Draw** event: for each cell, **If** the variable is not empty, **Draw text** it in the cell (add 100 to x per column and to y per row).

You should see: press **F5**. A white grid appears. Clicking a cell puts an X in it; clicking it again does nothing.

Phase 2: Host a Game / Join a Game

- ☐ **6.** Pick a folder both computers can use (a network drive, or a synced folder). For solo testing, a plain folder name like tictactoe_files is fine.
- ☐ **7. Key press H:** **Host a Game (File Exchange)** with that folder, max players 2, name "Player 1"; then set my_mark to X. **Key press J:** **Join a Game (File Exchange)** with the same folder and name "Player 2"; then set my_mark to O.
- ☐ **8.** In the **Draw** event, before the grid: **If** global.waiting_for_players == 1 then draw "Waiting for opponent..." and **Exit event**; otherwise draw "You are " + my_mark.

You should see: run the game on both computers. Press H on one and J on the other. "Waiting for opponent..." disappears and each screen says "You are X" or "You are O".

Phase 3: Taking Real Turns

- ☐ **9. Create:** set `last_round_acted` and `my_turn` to 0.
- ☐ **10. Step:** if it is my turn (`global.round_number` odd and I am X, or even and I am O) set `my_turn` to 1, otherwise 0. Then: if `my_turn == 0` and `last_round_acted != global.round_number`, *End Turn (File Exchange)* and set `last_round_acted` to `global.round_number` (a "pass").
- ☐ **11.** Rewrite the nine click checks: add `my_turn == 1`, and instead of setting the variable, *Set a Shared Variable (File Exchange)* with the cell's own name and the value `my_mark`, then *End Turn (File Exchange)*.
- ☐ **12.** In the **Draw** event, draw each cell from `global.cell_0_0` etc. (draw it only if it is not 0; an unset shared variable reads as 0).

You should see: only the player whose turn it is can place a mark. A mark shows on **both** screens a moment after the click, and turns alternate X, O, X.

Phase 4: Winning and Playing Again

- ☐ **13.** In `obj_game` add the **Round Resolved** event with eight **If** checks, one per line (three rows, three columns, two diagonals): if the three `global.cell_*` are equal and not 0, set winner to that mark.
- ☐ **14. Create:** set winner to an empty text. **Draw** (last): if `winner != ""`, draw `winner + " wins! Press SPACE to play again."`.
- ☐ **15. Key press Space:** if `winner != ""`, *Leave the Game (File Exchange)*. Press H and J again for a new game.

You should see: get three in a row and both screens show who won.

Stuck?

What is wrong	Most likely cause	What to do
The screen is empty	The drawing is black on the black room	Add *Set draw color* white at the top of the Draw event
It says "Waiting for opponent..." forever	The two computers use different folders, or a computer cannot reach the folder	Use exactly the same folder path; test that both can create a file there
Both computers say "You are X"	Both pressed H	One presses H, the other J
A click does nothing	It is not your turn, the cell is already used, or <code>my_turn</code> is not set	Wait for your turn; check the Step event
My mark shows only after a moment	That is normal: it appears after the round resolves (about a second)	Wait about a second
One player's mark never appears	They forgot *End Turn* after the click	Add *End Turn (File Exchange)* to the click handler

